



Computer Engineering Department

Course Name: Microprocessor Lab

Number: 10636392

Lab Report Grading Sheet

Instructor : Dr. Aladdin Masri	Experiment # : 9
Academic Year : 2025 - 2026	Experiment Name: Controlling a Stepper Motor
Semester : second semester	

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Performed on : 10/5/2026			Submitted on : 15/5/2026	
Report's Outcomes				
ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %	ILO __ =() %
Evaluation Criterion			Grade	Points
Abstract answers of the questions: “What did you do? How did you do it? What did you find?” Introduction and Theory Sufficient, clear and complete statement of objectives. In addition to Presents sufficiently the theoretical basis. Apparatus/ Procedure Apparatus sufficiently described to enable another experimenter to identify the equipment needed to conduct the experiment.Procedure sufficiently described.			0.5	
Experimental Results and Discussion (In-Lab Worksheet) Crisp explanation of experimental results. Comparison of theoretical predictions to experimental results, including discussion of accuracy and error analysis in some cases.			1.5	
Conclusions and Recommendations Conclusions summarize the major findings from the experimental results with adequate specificity. Recommendations appropriate in light of conclusions. Correct grammar.			2	
Appearance Title page is complete, page numbers applied, content is well organized, correct spelling, fonts are consistent, good visual appeal.			4	
			1	
			1	
Total			10	



Objectives

Gain a fundamental understanding of stepper motor control.

Develop proficiency in programming techniques to control stepper motor rotation.

Introduction

Stepper motors are DC motors that move in discrete steps. They have multiple coils that are organized in groups called "phases". By energizing each phase in sequence, the motor will rotate, one step at a time. By controlling the stepping of the motor, you can achieve very precise positioning and/or motor speed. For this reason, stepper motors are the motor of choice for many precision motion control applications.

Operating Modes:

1. full step (on one-phase): the motor is operated with only one phase energized at a time. This mode requires the least amount of power from the driver of any of the other modes.
2. Half step: it is a combination of one phase on and two phase on full step modes. This results in half the basic step angle. This smaller step angle provides smoother operation due to the increased resolution of the angle.



Procedure and Discussion

PART ONE: Rotate the stepper motor continuously For full step

```
1 ;FULL STEP
2 CODE SEGMENT
3 ASSUME CS:CODE
4 ORG 2000H
5
6 START:
7
8 mov dx,0ff2bh
9 mov al,80h
10 out dx,al
11
12 mov bl , 00010001b
13
14 lp:
15 mov dx,0ff28h
16 mov al,bl
17 out dx,al
18 ror bl, 1
19
20 mov cx , 0ffh
21 delayLp:
22 loop delayLp
23 jmp lp
24
25
26 END START
27 CODE ENDS
28
```

Half step

```
1 CODE SEGMENT
2 ASSUME CS:CODE
3 ORG 2000H
4
5 START:
6 mov dx,0ff2bh
7 mov al,80h
8 out dx,al
9
10 LP:
11 lea si,array
12 mov cx,8
13
14 LP2:
15 mov dx,0ff28h
16 mov al,[si]
17 out dx,al
18
19 push cx
20 mov cx,0ffh
21 delayLp:
22 loop delayLp
23 pop cx
24
25 inc si
26 loop LP2
27
28 jmp LP
29
30 array db 00000001b,00000011b,0000010b,00000110b,00000100b,00001100b,00001000b,00001001b
31 END START
32 CODE ENDS
33
```



Part 2 Code :

Use two push buttons to control the rotation of the stepper motor.

When btn1 is pressed, the motor rotates in a clockwise direction continuously,
and btn2 rotates the motor in a counterclockwise direction.

```

1  CODE SEGMENT
2  ASSUME CS:CODE
3  ORG 2000H
4
5  Start:
6  mov dx,0ff2bh
7  mov al,82h
8  out dx,al
9
10 mov bl , 00010001b
11
12 readButtons:
13 mov dx,0ff29h
14 in al,dx
15 or al,00111111b
16 cmp al,01111111b
17 je ClockWise
18 cmp al,10111111b
19 je CounterClockWise
20 jmp readButtons
21
22 ClockWise:
23 mov dx,0ff28h
24 mov al,bl
25 out dx,al
26 rol bl,1
27
28 push cx
29 mov cx,0ffffh
30 delay:
31 nop
32 loop delay
33 pop cx
34
35 jmp readButtons
36
37 CounterClockWise:
38 mov dx,0ff28h
39 mov al,bl
40 out dx,al
41 ror bl,1
42
43 push cx
44 mov cx,0ffffh
45 delay1:
46 nop
47 loop delay1
48 pop cx
49
50 jmp readButtons
51
52 END START
53 CODE ENDS
54

```



Part 2 Code :

PART THREE: Update Task 2 by rotating left the stepper motor a single rotation (150°) whenever a first button is pressed and rotating right the stepper motor a half rotation (90°) whenever a second button is pressed.

```

1 CODE SEGMENT
2 ASSUME CS:CODE
3 ORG 2000H
4
5 Start:
6 mov dx,0ff2bh
7 mov al,82h
8 out dx,al
9
10 mov bl , 00010001b
11
12 readButtons:
13 mov dx,0ff29h
14 in al,dx
15 or al,00111111b
16 cmp al,01111111b
17 je ClockWise
18 cmp al,10111111b
19 je CounterClockWise
20 jmp readButtons
21
22 ClockWise:
23 mov cx, 852
24 lp:
25     mov dx,0ff28h
26     mov al,bl
27     out dx,al
28     rol bl,1
29     push cx
30     mov cx,0ffffh
31     delay:
32     nop
33     loop delay
34     pop cx
35 loop lp
36 jmp readButtons
37
38 CounterClockWise:
39 mov cx, 511
40 lp2:

```

```

21
22 ClockWise:
23 mov cx, 852
24 lp:
25     mov dx,0ff28h
26     mov al,bl
27     out dx,al
28     rol bl,1
29     push cx
30     mov cx,0ffffh
31     delay:
32     nop
33     loop delay
34     pop cx
35 loop lp
36 jmp readButtons
37
38 CounterClockWise:
39 mov cx, 511
40 lp2:
41     mov dx,0ff28h
42     mov al,bl
43     out dx,al
44     ror bl,1
45     push cx
46     mov cx,0ffffh
47     delay2:
48     nop
49     loop delay2
50     pop cx
51 loop lp2
52 jmp readButtons
53
54 END START
55 CODE ENDS
56

```



Conclusion

Stepper motors are moved in discrete steps, and there are different ways they can do it. In "full step," they take bigger, stronger strides. In "half step," they take smaller, smoother steps. Then there's "micro stepping," where they move in tiny, precise increments. Each mode has its use – whether it's about power, smoothness, or pinpoint accuracy. These motors are crucial in devices that need controlled and accurate motion, like printers or machines in factories.